

Markov's inequality



$$X = \text{random variable} \geq 0 \\ \text{any real value } a > 0 : \Pr[X \geq a] \leq \frac{E[X]}{a}$$

R bounds a prob. with an expectation

proof

Use an indicator variable $I = \begin{cases} 1 & \text{if } X \geq a \\ 0 & \text{o.w. (X < a)} \end{cases}$

$$E[I] = 0 \cdot \Pr[I=0] + 1 \cdot \Pr[I=1] = \Pr[X \geq a] \quad (*)$$

claim: $aI \leq X$

- $X \geq a \Rightarrow I=1 \Rightarrow aI = a \leq X$
- $X < a \Rightarrow I=0 \Rightarrow aI = 0 \leq X$

$$\left. \begin{array}{l} E[aI] \leq E[X] \\ \text{"} \\ a E[I] \stackrel{(*)}{=} a \Pr[X \geq a] \end{array} \right\} \Rightarrow a \Pr[X \geq a] \leq E[X]$$

k -wise limited independence

$X_1, X_2, \dots, X_n$ random variables

► k -wise independent of

$\forall$ choice $1 \leq i_1 < i_2 < \dots < i_k \leq n$ and values $a_1, a_2, \dots, a_k$

$$\Pr[X_{i_1} = a_1 \wedge X_{i_2} = a_2 \wedge \dots \wedge X_{i_k} = a_k] = \Pr[X_{i_1} = a_1] \times \Pr[X_{i_2} = a_2] \times \dots \times \Pr[X_{i_k} = a_k]$$

We use this notion for a family of hash functions $\mathcal{H}$

In particular, we need the case $k=2$ for universal hashing

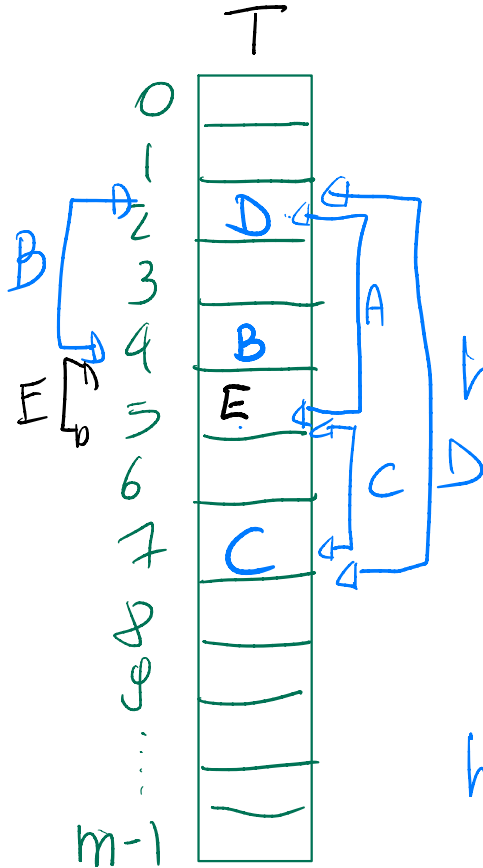
$$h \in \mathcal{H} \quad h(x) = ((ax + b) \% p) \% m$$

$\forall$ choice $i, j \in [m]$ and $x \in U$

$$\Pr_{h_1, h_2 \in \mathcal{H}} [h_1(x) = i \wedge h_2(x) = j] = \Pr_{h_1 \in \mathcal{H}} [h_1(x) = i] \times \Pr_{h_2 \in \mathcal{H}} [h_2(x) = j] \sim \frac{1}{m^2}$$

CUCKOO HASHING

- it guarantees $O(1)$ worst-case lookup as perfect hash but using just two hash functions $h_1, h_2 \in \mathcal{H}$
- it can be also updated in $O(1)$ amortized expected time



x	A	B	C	D	E
$h_1(x)$	2	2	5	7	4

ISSUE $\Rightarrow$ CYCLE

x	A	B	C	D	E
$h_2(x)$	5	4	7	2	5

Lookup(x) : check if x is found in either $T[h_1(x)]$ or $T[h_2(x)]$
Delete(x) : remove x if it is found in $T[h_1(x)]$ or $T[h_2(x)]$ } $O(1)$ worst case time

Insert(x): $O(1)$ expected time

→ if $T[h_1(x)]$ or $T[h_2(x)]$ is empty, store x in that entry of T
else let y be the key in one of these occupied entries of T
(y is now homeless)
x is stored in place of y, and y becomes the new x

FAILING CONDITION to check here:

when the table is full or a cyclic situation occurs

(just check if #iterations > # stored key)

[see Python code]

FAILING PROCEDURE: choose again $h_1, h_2 \in \mathcal{H}$ randomly and uniformly
and rebuild everything from scratch

We prove that $\text{Insert}(x)$ takes $O(i)$ expected time
(amortized)

$S = \text{set of } n \text{ keys}$
 $h_1, h_2 \in H$

$\Rightarrow$ undirected graph $G = (V, E)$
 $V = \{0, 1, \dots, m-1\}$
 $E = \{h_1(x)h_2(x) : x \in S\}$ (actually G is a multigraph)

In G , we can trace the insertion of x as a path from node i to node j
 $i, j \in [m]$

In particular, inserting x corresponds to adding edge $h_1(x)h_2(x)$

Three possible situations occur:

① x finds soon an empty entry $O(1)$ time

② the positions $h_1(x)$ and $h_2(x)$ are already taken
insertion follows a path in G that starts from $i \in \{h_1(x), h_2(x)\}$
and ends in j

$$O(1 + \ell)$$

$\ell = \#$ traversed
edges (an edge can
be traversed
more than
once)

③ like ②, but the path contains a cycle
 $\Rightarrow$ REHASHING

Important: $m > 2cn$ for a constant $c > 2$
we focus on cases ② and ③

case (2)

$$m > 2cw$$

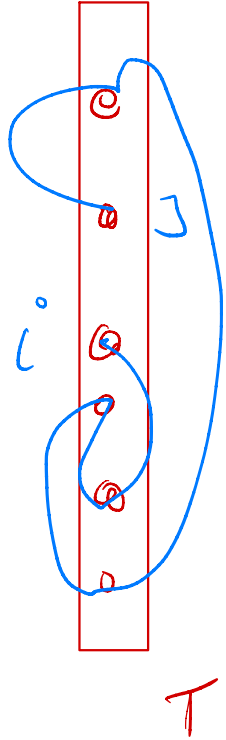
(X) [we prove by induction on $\ell \geq 1$ that
probability that path length is $\ell \leq \frac{1}{c^\ell m}$

[since the insertion cost is $O(1+\ell)$ time
we obtain an expected cost

$$O\left(1 + \sum_{\ell \geq 1} \left[\ell \cdot \frac{1}{c^\ell m} \right] \right) = O\left(1 + \frac{1}{m}\right) = O(1) \text{ expected time}$$

Knuth, Concrete Math, p. 33

$$\sum_{\ell \geq 1} \ell \left(\frac{1}{c}\right)^\ell < \frac{c}{(c-1)^2} \quad c > 2$$



(X) we will prove it in the next class

case ③

since a cycle appears in G , a REHASHING is performed in $O(n)$ time

$$\begin{aligned}\Pr(\exists \text{ cycle in } G) &= \sum_{i=0}^{m-1} \Pr(\exists \text{ path from } i \text{ to } j=i) \\ &= \sum_{i=0}^{m-1} \sum_{l \geq 1} \underbrace{\Pr(\exists \text{ path from } i \text{ to } j \text{ of length } l)}_{\leq \frac{1}{c^l m}}\end{aligned}$$

$$\leq \sum_{i=0}^{m-1} \sum_{l \geq 1} \frac{1}{c^l m} = \sum_{l \geq 1} \frac{1}{c^l} \leq \frac{1}{c-1} = p \quad \left[\begin{array}{l} c \geq 2 \\ c \geq 2 \end{array} \right]$$

Q. How many rehashings?

1 with p

2 with p^2

$\vdots$ with p^i

$\Rightarrow$ the expected number of rehashings

$$\text{is } \sum_{i \geq 1} i p^i = O(1) \text{ as } p < 1$$

HANDS-ON

- ① Prove that deletion does not leave necessarily a random graph
[$n=3$ is enough]
- ② Show how to fix ①
- ③ Prove $\Pr(\text{path of length } e) \leq \frac{1}{c^e m}$

$$\begin{array}{c} i \xrightarrow{e-1} k \xrightarrow{1} A \xrightarrow{1} \frac{1}{c} \\ \text{induction} \leq \frac{1}{c^{e-1} m} \end{array}$$

